Project Planning & Management Document

1. Project Overview

Project Title:
Line-Following Emotion Robot

Team Members & Roles:

- **Ishanvi** — OLED programming, maze fabrication, robot design and structure assembly, MIT App Inventor app development.
- **Manasvi** — Final coding and all other iterations, circuit assembly, robot design and structure assembly.

Brief Description:
This project is an interactive line-following robot built using ESP32 and MicroPython. The final system combines line tracking, color detection, and an OLED-based facial display to create a robot that not only moves along a path but also expresses different emotions based on detected colors.

2. Team Roles & Responsibilities

Member Name	Role	Responsibilities
Ishanvi	Display, fabrication, app support	Wrote the OLED display code; built the entire maze for the original concept; helped assemble the robot body, design, and structure; developed the MIT App Inventor app.
Manasvi	Programming and electronics	Wrote all the test codes, all iterations, final code, BLE code, made all test circuits, final circuit, all circuit iterations. Joined the entire robot chassis structure. Helped with art and crafts.

The responsibilities were divided based on strengths in coding, fabrication, and hardware integration. This distribution allowed the project to progress in parallel, with one member focusing more on display and physical construction while the other handled the final software logic and electronics integration.

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 1	Finalize concept, begin hardware testing, start movement setup	Project concept, tested drivers, initial motor setup	Completed
Week 2	Continue maze-navigation concept, improve robot body and circuit	Assembled robot structure, partial navigation testing	Completed
Week 3	Integrate maze system and prepare for final testing	Maze completed, robot tested in navigation setup	Completed
Week 4	Solve final hardware failure, redesign concept, complete final system	Line-following emotion robot, final code, completed submission	Completed

4. Weekly Progress Log

Week 1

Planned Tasks:
Finalize the concept, begin motor testing, assemble the basic electronics, and start prototyping the robot.

Completed Tasks:
The initial concept selected for the project was a **self maze-navigating robot**. Basic circuit setup began, and multiple motor drivers were tested in order to achieve reliable movement.

Challenges Faced:
For almost the entire first week, the project was delayed because every motor driver tested turned out to be faulty. A total of **six motor drivers** had to be replaced before a usable one was found.

Solutions:
Each driver was systematically tested, replaced, and rewired until the movement system responded correctly. This process helped identify that the issue was not with the motors or code, but with the drivers themselves.

Next Steps:
Once a functional driver was available, the next goal was to continue building the robot and test movement for maze navigation.

Week 2

Planned Tasks:
Develop the robot structure, continue assembling the hardware, and begin testing the maze-navigation concept.

Completed Tasks:
The physical robot structure was assembled further, and work continued on the original maze-solving idea. The circuit layout and body design were adjusted to fit the movement and sensing components more effectively.

Challenges Faced:
The project had already lost time due to hardware faults in the first week, which placed pressure on the remaining development timeline. Movement and alignment also required repeated adjustment.

Solutions:
The chassis layout was refined, and the robot body was assembled more carefully to support stability. Tasks were divided clearly so that coding, fabrication, and assembly could happen in parallel.

Next Steps:
Continue refining the maze robot, complete testing in the fabricated maze, and prepare the system for full integration.

Week 3

Planned Tasks:
Complete the maze, test the self maze-navigating robot, and stabilize the integrated system.

Completed Tasks:
The maze for the original concept was completed, and the robot was tested within that setup. Both the physical structure and electronics were refined to support the navigation-based concept.

Challenges Faced:
Even after assembly, the robot still needed reliable movement consistency. The project

remained dependent on motor performance, which continued to be a sensitive point in the system.

Solutions:
Further small corrections were made in component placement, assembly, and coding. The team continued testing the system with the expectation of completing the maze-navigation concept.

Next Steps:
Conduct final trials and finish the original concept in time for submission.

Week 4

Planned Tasks:
Carry out final testing, make final corrections, and prepare for submission.

Completed Tasks:
A major last-minute change was made to the project. **One day before submission, one of the motors became faulty**, which made the self maze-navigating robot concept unreliable. Because of this, the project was **redesigned overnight** into the final **line-following emotion robot**. The final code was written, the circuit was reassembled where needed, and the updated system was tested in its new format.

Challenges Faced:
The motor failure occurred at the most critical stage of the project and made the original concept impossible to complete with confidence. This required rapid decision-making under time pressure.

Solutions:
Instead of attempting to force the original concept to work, the project scope was changed strategically to a more achievable and stable model. The robot was adapted into a line-following system, and the emotional OLED interaction became the core feature of the final submission.

Next Steps:
Complete the documentation, finalize the report, and prepare the project for final evaluation.

5. Reflection on Project Management

The project required continuous adaptation due to repeated hardware issues and last-minute changes. While the original plan was to create a self maze-navigating robot, technical failures in both the motor drivers and the motor itself forced a shift in direction. This made flexibility one of the most important parts of the team’s management process.

The division of responsibilities helped the team respond quickly to these setbacks. By separating coding, fabrication, app support, and circuit integration across team members, it became easier to continue progress even when one area encountered problems. The final outcome demonstrates not only technical integration but also the team’s ability to manage unexpected challenges within a limited time frame.